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Assignment 2

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 108 = 3 + 105 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

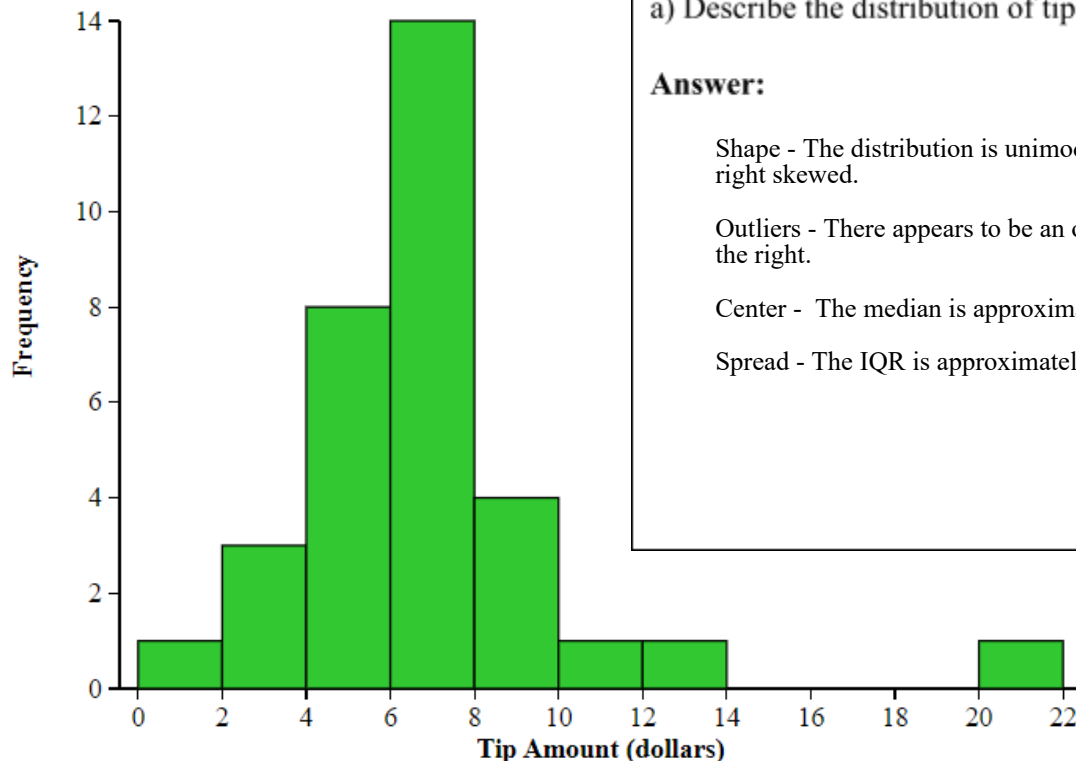
Reflective Journal [3 pts]

(link to your live google doc)

<https://docs.google.com/document/d/1dzAggy7UhyrmR9CyIVDsXdToMj1JATrVwGZaqPQtogo/edit?usp=sharing>

I. SOCS and Comparing (33 pts = 11 x 3 pts)

1) Karley works as a Dominoes delivery driver and on Friday night, she recorded how many tips she got from each of her deliveries. The histogram below shows the distribution of the 33 deliveries.



a) Describe the distribution of tips.

Answer:

Shape - The distribution is unimodal and right skewed.

Outliers - There appears to be an outlier on the right.

Center - The median is approximately 6-7.

Spread - The IQR is approximately 2-3.

b) One of the tip amounts was \$9. If this tip had been \$14 instead of \$9, describe what effect this would have on the following measures of the center. Justify your answer.

Mean:

The mean would increase because the sum of tips would increase but the number of tips wouldn't change.

Median:

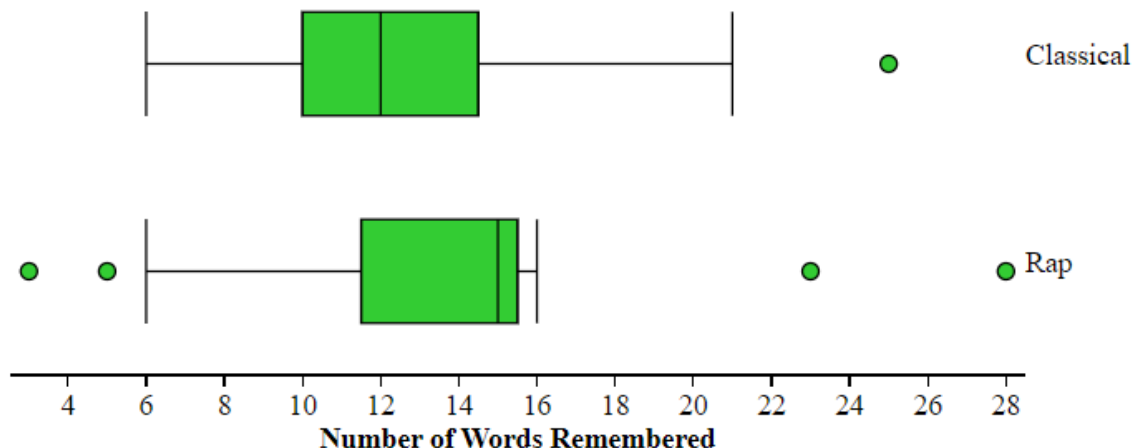
The median wouldn't be effected. The median appears to be less than \$9, so increasing the value of a tip that is greater than the median won't change the number of tips on either side of the center so the median won't change.

c) Describe how you would estimate the median of this distribution.

Answer:

The exact values of the tips are unknown, but they can be estimated as the mean of the bin in the histogram. Then the number of tips in each bin can be counted to determine the center of the distribution to estimate the median.

2) Jordan and Alex want to know if listening to different types of music while studying will help you remember the material better. They randomly assigned a group of students to two different groups: one group studied a list of words while listening to classical music while the second group studied the same list of words while listening to rap music. There was a total of 30 words on the list and each group studied the list while listening to the music for 5 minutes. They were then asked to write down as many words as they could remember. The data is displayed in the boxplots below.



a) Approximate the interquartile range for each set of data. Why is this the appropriate measure of spread to use for these two data sets?

Answer:

The interquartile range for classical music is approximately 4.5 words.
The interquartile range for rap music is approximately 4 words.
Interquartile range is the appropriate measure because there are outliers which would strongly influence other measures of spread.

b) Write three sentences comparing the number of words remembered between each of the two groups.

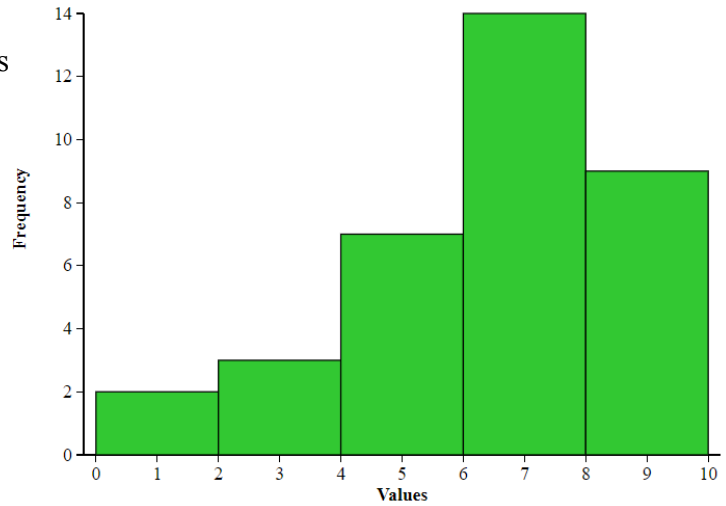
Answer:

The median number of words recalled is higher for students who listened to rap than classical.
The distribution of words remembered is roughly symmetric for students who listened to classical music but skewed left for students who listened to rap.
There are more outliers among the group of students who listened to rap than the group of students who listened to classical.

Multiple Choice Practice

A

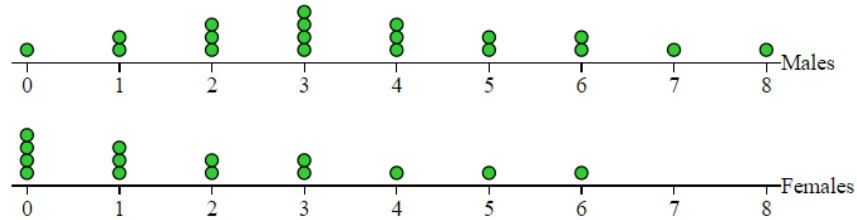
3) The histogram below displays a set of measurements. Which of the boxplots below displays the same set of measurements?



- (A)
- (B)
- (C)
- (D)
- (E)

B

4) These dotplots for randomly selected male and female students at a particular high school show the number of times per week they eat at fast food restaurants.



Which of the following is a true statement?

- (A) One distribution is roughly symmetric; the other is skewed left.
- (B) The males' median is larger than the females' mean.
- (C) The ranges are both 8.
- (D) The standard deviations are equal.
- (E) Combining the male and female times into one set of student times will increase the range to 16.

D

5) The boxplots below summarize two sets of data, A and B. Which of the following must be true?



- I. Set A contains more data than Set B.
- II. Set A has a larger range than Set B.
- III. Set A and Set B have the same median.

- (A) I only
- (B) III only
- (C) I and II only
- (D) II and III only
- (E) I, II, and III

B

6) Which of the following statements is incorrect?

- (A) Both dotplots and stemplots can show symmetry, gaps, clusters, and outliers.
- (B) Sets with different distribution shapes can have identical boxplots.
- (C) Boxplots, dotplots, stemplots, and histograms can all show skewness.
- (D) In histograms, relative areas correspond to relative frequencies.
- (E) In histograms, frequencies can be determined from relative heights.

E

7) Consider the following test scores:

Class A: 65, 72, 73, 73, 74, 82, 91

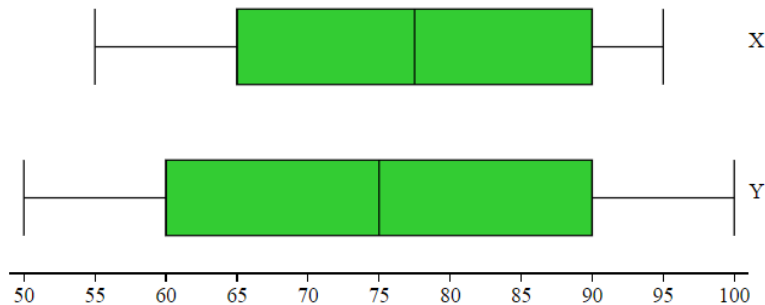
Class B: 60, 65, 79, 81, 85, 86, 86

Which of the following is a true statement?

- (A) Class A has the greater range, while Class B has the greater standard deviation.
- (B) Class B has the greater range, while Class A has the greater standard deviation.
- (C) The ranges and standard deviations are the same for both classes.
- (D) The ranges are the same, but Class A has the higher standard deviation.
- (E) The ranges are the same, but Class B has the higher standard deviation.

C

8) The boxplots shown below summarize two sets of data, X and Y.



Which of the following must be true?

- (A) 50 is an outlier for Set Y.
- (B) Set Y contains more data than Set X.
- (C) The IQR of Set X is smaller than the IQR of Set Y.
- (D) The range of Set Y is smaller than the range of Set X.
- (E) The means of the two data sets are equal.

II. The Empirical Rule and Z-Scores (72 pts = 24 x 3 pts)

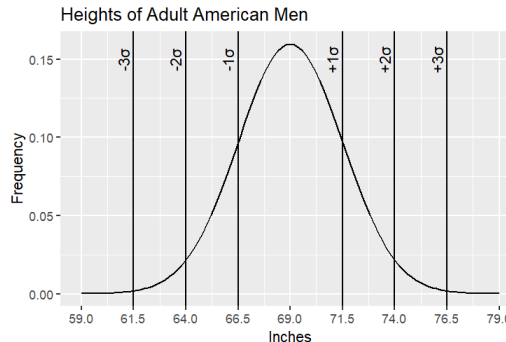
1) Larry came home very excited after a visit to his doctor. He announced proudly to his wife, “My doctor says my blood pressure is at the 90th percentile among men like me. That means I’m better off than about 90% of similar men.” How should his wife, who is a statistician, respond to Larry’s statement?

Answer:

Larry correctly understands that 90th percentile means that he is above 90% of the population. However, he is not understanding that having a blood pressure higher than 90% of similar men is a bad thing. His wife should explain to him that for measures of health such as blood pressure, it's better to be within the normal range than to be on the tails of the distribution.

2) The distribution of heights of adult American men is approximately normal with mean 69 inches and standard deviation 2.5 inches. Draw a normal curve below and mark three standard deviations out on either side. Then, use the curve to answer the following questions.

Answer:



(a) What percent of men are shorter than 64 inches?

Using the empirical rule, we know that 95% of data fall within 2 standard deviations of the mean. This means 5% of data don't fall within 2 standard deviations, so 2.5% are below -2 standard deviations and 2.5% are above +2 standard deviations. Since 64 inches is -2 standard deviations below the mean, there are 2.5% of men shorter than 64 inches.

(b) What percent of men are taller than 71.5 inches?

Using the empirical rule, we know that 68% of data fall within 1 standard deviation of the mean. This means 32% of data don't fall within 1 standard deviation, so 16% are below -1 standard deviation and 16% are above +1 standard deviation. Since 71.5 inches is +1 standard deviation above the mean, there are 16% of men taller than 71.5 inches.

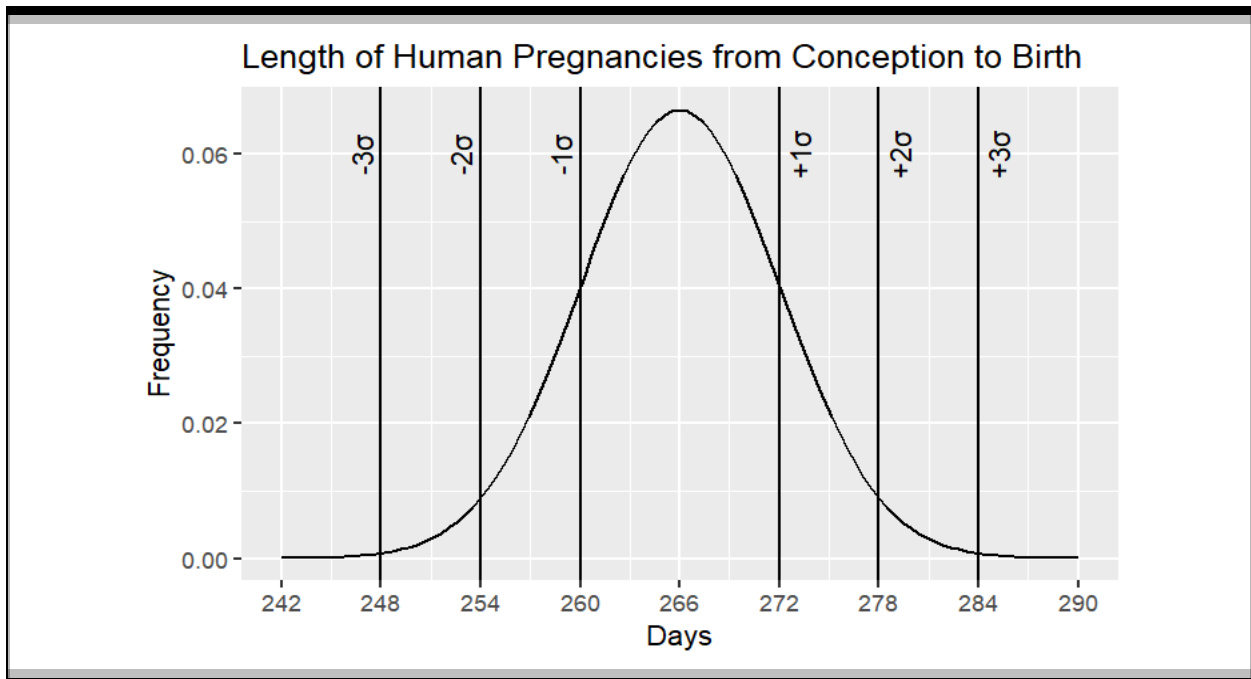
(c) What percent of men are taller than 66.5 inches?

Using the empirical rule, we know that 68% of data fall within 1 standard deviation of the mean. This means 32% of data don't fall within 1 standard deviation, so 16% are below -1 standard deviation and 16% are above +1 standard deviation. Since 66.5 inches is -1 standard deviation below the mean, there are $68\% + 16\% = 84\%$ of men taller than 66.5 inches.

(d) A height of 74 inches corresponds to what percentile of American male heights?

Using the empirical rule, we know that 95% of data fall within 2 standard deviations of the mean. This means 5% of data don't fall within 2 standard deviations, so 2.5% are below -2 standard deviations and 2.5% are above +2 standard deviations. Since 74 inches is +2 standard deviations below the mean, there are $95\% + 2.5\% = 97.5\%$ of men shorter than 74 inches. This corresponds to the 97.5th percentile at 74 inches.

3) The length of human pregnancies from conception to birth varies on an approximate normal distribution with mean 266 days and a standard deviation of 6 days. Draw a normal curve below and mark three standard deviations out on either side. Then, use the curve to answer the following questions.



a) How long are the longest 16% of pregnancies?

The longest 16% of pregnancies are above 1 standard deviation from the mean, according to the empirical rule. Therefore, the longest 16% of pregnancies are longer than 272 days.

b) How short are the shortest 16% of pregnancies?

The shortest 16% of pregnancies are below 1 standard deviation from the mean, according to the empirical rule. Therefore, the shortest 16% of pregnancies are shorter than 260 days.

c) Between how many days does the middle 95% of all pregnancies fall?

According to the empirical rule, 95% of data falls within 2 standard deviations. Therefore, the middle 95% of pregnancies fall within 254 - 278 days.

d) Between how many days does the middle 99.7% of all pregnancies fall?

According to the empirical rule, 99.7% of data falls within 3 standard deviations. Therefore, the middle 99.7% of pregnancies fall within 248 - 284 days.

e) A pregnancy of 254 days corresponds to what percentile of pregnancies?

Using the empirical rule, we know that 95% of data fall within 2 standard deviations of the mean. This means 2.5% are below -2 standard deviations and 2.5% are above +2 standard deviations. Since 254 days is -2 standard deviations below the mean, there are 2.5% of pregnancies shorter than 254 days, which is the 2.5th percentile.

4) Three landmarks of baseball achievement are Ty Cobb's batting average of 0.420 in 1911, Ted Williams's 0.406 in 1941, and George Brett's 0.390 in 1980. These batting averages cannot be compared directly because the distribution of major league batting averages has changed over the years. The distributions are quite symmetric, except for outliers such as Cobb, Williams, and Brett. While the mean batting average has been held roughly constant by rule changes and the balance between hitting and pitching, the standard deviation has dropped over time. Here are the facts:

Decade	Mean	Standard Deviation
1910s	0.266	0.0371
1940s	0.267	0.0326
1970s	0.261	0.0317

Find the standardized scores for Cobb, Williams, and Brett. Who was the best hitter?

Cobb: $(0.420 - 0.266) / 0.0371 = 4.15$

Williams: $(0.406 - 0.267) / 0.0326 = 4.26$

Brett: $(0.390 - 0.261) / 0.0317 = 4.07$

Ted Williams was the best hitter because he has the highest standardized score.

5) The scores of an introductory statistics class at the University of Wisconsin is normally distributed with a mean of 72 and standard deviation of 10.

(a) If a student has a z-score of 1.4, what score does she have?

$$\begin{aligned} Z \text{ score} &= (x - \mu) / \sigma \\ 1.4 &= (x - 72) / 10 \\ 14 &= x - 72 \\ 86 &= x \\ \text{The student has a score of 86.} \end{aligned}$$

(b) If a student has a z-score of -1.23, what score does he have?

$$\begin{aligned} Z \text{ score} &= (x - \mu) / \sigma \\ -1.23 &= (x - 72) / 10 \\ -12.3 &= x - 72 \\ 59.7 &= x \\ \text{The student has a score of 59.7.} \end{aligned}$$

6) A college statistics exam scores were normally distributed with $\mu = 150$ points and $\sigma = 20$ points. The z-scores for some students are below. Find the final exam raw score for each.

Chris 0.1

$$\begin{aligned} Z \text{ score} &= (x - \mu) / \sigma \\ 0.1 &= (x - 150) / 20 \\ 2 &= x - 150 \\ 152 &= x \\ \text{Chris has a score of 152.} \end{aligned}$$

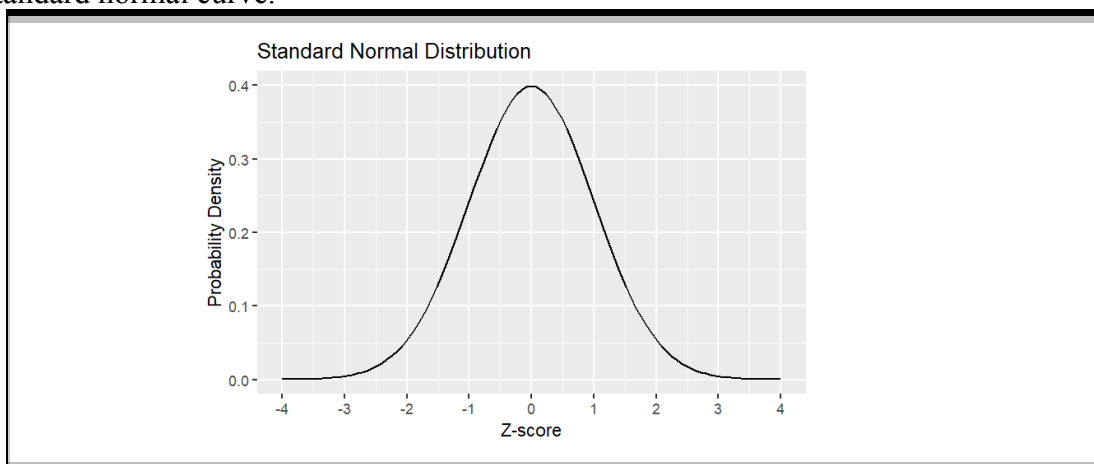
Janet 1.8

$$\begin{aligned} Z \text{ score} &= (x - \mu) / \sigma \\ 1.8 &= (x - 150) / 20 \\ 36 &= x - 150 \\ 186 &= x \\ \text{Janet has a score of 186.} \end{aligned}$$

Rick -1.1

$$\begin{aligned} Z \text{ score} &= (x - \mu) / \sigma \\ -1.1 &= (x - 150) / 20 \\ -22 &= x - 150 \\ 128 &= x \\ \text{Rick has a score of 128.} \end{aligned}$$

7) Make an accurate sketch of a Standard Normal Distribution and answer each question using a standard normal curve.



a) What is the area between $z = 0$ and $z = 1$?

Using the empirical rule, 68% of data falls within 1 standard deviation. The area between $z = 0$ and $z = 1$ is exactly half of the area between $z = -1$ and $z = 1$. Therefore, the area between $z = 0$ and $z = 1$ is 0.34. The exact value when looked up in a z-score table is 0.3413.

b) What percent of the data is within one standard deviation of the mean?

Using the empirical rule, 68% of data falls within 1 standard deviation of the mean. Using the z-score table it is twice the area between $z = 0$ and $z = 1$ because the distribution is perfectly symmetric. Therefore the exact value is $0.3413 * 2 = 0.6826$ or 68.26%.

c) What percent of the data is more than one standard deviation from the mean?

The area under the curve is exactly 1, and from the previous question the area within one standard deviation is 0.6826. Therefore, the area outside one standard deviation is 0.3174 and the percent of data more than one standard deviation from the mean is 31.74%.

d) What is the area between $z = 0$ and $z = 2$?

Using the empirical rule, 95% of data falls within 2 standard deviations. The area between $z = 0$ and $z = 2$ is exactly half of the area between $z = -2$ and $z = 2$. Therefore, the area between $z = 0$ and $z = 2$ is 0.475. The exact value when looked up in a z-score table is 0.4772.

e) What is the area under the curve between $z = -1$ and $z = 2$?

Using the answers from the previous questions, the area under the curve from $z = 0$ to $z = 1$ is 0.3413 and from $z = 0$ to $z = 2$ is 0.4772. Because the curve is perfectly symmetric, the area from $z = 0$ to $z = 1$ is exactly the same as from $z = -1$ to $z = 0$. Therefore, the area under the curve between $z = -1$ and $z = 2$ is equal to the sum of the areas under the curve from $z = 0$ to $z = 1$ and $z = 0$ to $z = 2$. The exact value of area under the curve between $z = -1$ and $z = 2$ is $0.3413 + 0.4772$ which is equal to 0.8185.

THE END